Assignment 7: Orbital Dynamics II: Interacting Systems

PHYS 4202 - Computational Physics (Fall 2026)
Due: Tuesday, October 13, 2026 at 10:00 am
Total points: 150 + 20 extra credit

Purpose

In Assignment 6, the central mass was fixed and only one object moved. In this assignment, every body responds to the gravitational pull of every other body. You will extend Velocity Verlet to an interacting N-body system, validate it with two-body systems, and study a three-body encounter that produces an ejection. An optional extra-credit part investigates a long-lived rotating three-body configuration.

The emphasis is on:

- simultaneous force evaluation
- total energy and momentum diagnostics
- center-of-mass behavior
- (in)stability of multi-object interactions

Environment

All work must be done in **Python 3.9 or newer**. You may use the Python standard library, NumPy, and matplotlib. You must implement the integrator yourself.

Code organization

Your code directory must contain the following Python files:

- `body.py`: defines the `Body` class.
- `simulation.py`: contains pairwise acceleration, Velocity Verlet, and diagnostic functions.
- `binary_orbit.py`: runs the comparable-mass simulations and generates the Part 2 figures.
- `mass_ratio_limit.py`: runs the large-mass-ratio simulations and generates the Part 3 figures.
- `three_body_ejection.py`: runs the three-body simulations and generates the Part 4 figures.

If you complete the extra credit, also include `lagrange_orbit.py`.

You may create additional Python modules and import them into the required files if they help you organize and structure your code. These modules are entirely optional. Make sure to include any additional modules that your required files use in your git repository so your code runs as intended.

The `Body` class must store mass, current position and velocity, and the position and velocity histories. A method that records the current state is appropriate. Force calculations, integration, system diagnostics, and plotting should be implemented as external functions because each depends on more than one body.

Type hints are encouraged but not required.

All plots must have labeled axes, units where appropriate, and a legend whenever more than one data set or reference feature appears. Use equal physical scales on the x and y axes of every trajectory plot to ensure no aspect-ratio distortion. Include the required figures in your report. Do not submit figures separately.

Pairwise gravitation

The displacement from body i to body j is

$$\mathbf{d}_{ij} = \mathbf{r}_j - \mathbf{r}_i \qquad r_{ij} = \|\mathbf{d}_{ij}\|$$

The gravitational force on body i due to body j is

$$\mathbf{F}_{ij} = G m_i m_j \frac{\mathbf{d}_{ij}}{r_{ij}^3}$$

The total acceleration for body i is the sum of $\mathbf{F}_{ij}/m_i$ for all other bodies j.

The distance r_{ij} is the distance between the two bodies. It is not the distance of either body from the origin. The origin is an arbitrary choice and has no physical role in an interacting system.

Simultaneous Updates

Although General Relatively provides a finite speed of propagation for gravity, Newtonian gravity treats gravitational influence as propagating infinitely fast. Therefore, the accelerations of all objects within a gravitational system should be calculated from the same snapshot of position. Do not advance one body and then use its new position to calculate another body's motion for the same timestep. Record the state of all bodies only after the complete system has been advanced to the new time.

System Energy

Gravity is a conservative force, meaning that the total energy of an orbital system does not change. The energy of any individual object within the system can change and energy can change between kinetic energy and potential energy, but the total energy will remain constant. Your numerical solution should therefore be evaluated using the total system energy, not by expecting any individual contribution to remain constant.

For a system of interacting bodies, the total kinetic energy is

$$K = \sum_i \frac{1}{2} m_i \|\mathbf{v}_i\|^2$$

The total gravitational potential energy can be written as the double sum

$$U = -\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{G m_i m_j}{r_{ij}}$$

For each body i, the inner sum includes only bodies with $j > i$. Each distinct pair therefore appears exactly once, so there is no need to account for double counting. The total energy of the system is

$$E = K + U$$

For the required plots, assign half of each pair's potential energy to each body. Define

$$K_i = \frac{1}{2} m_i \|\mathbf{v}_i\|^2 \qquad U_i = -\frac{1}{2} \sum_{j \neq i} \frac{G m_i m_j}{r_{ij}} \qquad E_i = K_i + U_i$$

With this convention, $K = \sum_i K_i$, $U = \sum_i U_i$, and $E = \sum_i E_i$. The individual $K_i(t)$, $U_i(t)$, and $E_i(t)$ can vary even when the total system energy is conserved.

Center of Mass and Momentum

The total momentum of a system of moving objects is

$$\mathbf{p} = \sum_i m_i \mathbf{v}_i$$

The center-of-mass position and velocity of the system are

$$\mathbf{R}_{\text{CM}} = \frac{\sum_i m_i \mathbf{r}_i}{\sum_i m_i} \qquad \mathbf{V}_{\text{CM}} = \frac{\sum_i m_i \mathbf{v}_i}{\sum_i m_i}$$

All simulations should be performed in the center of mass frame to avoid systems drifting due to the frame of reference choice. To do this, write a function that takes a list of `Body` objects and updates the position and velocity of every body to the center-of-mass frame using

$$\mathbf{r}_i \longrightarrow \mathbf{r}_i - \mathbf{R}_{\text{CM}} \qquad \mathbf{v}_i \longrightarrow \mathbf{v}_i - \mathbf{V}_{\text{CM}}$$

This function will only need to be called once at the beginning of the simulation to adjust the initial states. For each simulation, you only need to begin with the relative velocities of each object and then use the function to adjust the values to ensure the full simulation is done in the center of mass frame. As long as all objects are updated simultaneously, there should be no drift observed.

For Parts 2–4, you will need to confirm that the total momentum and center-of-mass position do not change over time. To do this, you will need to provide a plot of

$$||\mathbf{R}_{CM}(t) - \mathbf{R}_{CM}(0)||$$

where $\mathbf{R}_{CM}(0)$ is the position of the initial center of mass. Make sure the y-axis is zoomed in sufficiently to show any potential deviations from zero.

Part 1: Body Class and N-Body Velocity Verlet (15 points)

In `body.py`, implement the `Body` class described above. In `simulation.py`, implement reusable functions that calculate all accelerations from the current body positions and advance every body by one Velocity Verlet step. Also implement the energy, momentum, and center-of-mass diagnostics defined above.

For one timestep, use the following order:

1. Calculate all accelerations $\mathbf{a}_i(t)$ from the complete set of current positions.
2. Calculate all next-step positions $\mathbf{r}_i(t+h)$.
3. Calculate all accelerations $\mathbf{a}_i(t+h)$ from the complete set of new positions.
4. Calculate all next-step velocities $\mathbf{v}_i(t+h)$ using both accelerations.
5. Record the new state of every body.

The physical studies in the later parts provide the principal tests of this implementation.

Part 2: Comparable-Mass Binary (35 points)

Simulate two bodies with

$$m_1 = 1, \qquad m_2 = 2.$$

Use an initial separation of $d = 1$. One suitable set of preliminary positions and velocities for an elliptical configuration is

$$\mathbf{r}_1(0) = (0,0) \qquad \mathbf{r}_2(0) = (1,0)$$
$$\mathbf{v}_1(0) = (0,-1.4) \qquad \mathbf{v}_2(0) = (0,0)$$

Be sure to use your center-of-mass-frame function to update both bodies before recording their initial states or beginning the simulation.

The relative position and velocity are

$$\mathbf{r}_{\text{rel}} = \mathbf{r}_2 - \mathbf{r}_1 \qquad \mathbf{v}_{\text{rel}} = \mathbf{v}_2 - \mathbf{v}_1$$

Let $v_0 = \|\mathbf{v}_{\text{rel}}(0)\|$. The semimajor axis of the relative orbit and the expected period are

$$\frac{1}{a_{\text{rel}}} = \frac{2}{d} - \frac{v_0^2}{G(m_1+m_2)} \qquad T = 2\pi\sqrt{\frac{a_{\text{rel}}^3}{G(m_1+m_2)}}$$

Calculate the expected period and simulate at least 10 complete elliptical orbits.

Each body follows its own ellipse about the center of mass. The two ellipses have the same eccentricity and share the center of mass as a focus. If $r_{i,\text{p}}$ and $r_{i,\text{a}}$ are the periapsis and apoapsis distances of body i from the center of mass, then

$$a_i = \frac{r_{i,\text{a}} + r_{i,\text{p}}}{2} \qquad e_i = \frac{r_{i,\text{a}} - r_{i,\text{p}}}{r_{i,\text{a}} + r_{i,\text{p}}}$$

- Include a trajectory plot showing both bodies, their initial positions, and the center of mass.
- Include a time-history plot of $K(t)$ for each body, $U(t)$ for each body, and $E(t)$ for each body as well as the whole system.
- Measure the period for each body and compare them with each other and the expected value.
- Measure the periapsis and apoapsis distances for both bodies. Calculate a_i and e_i for each orbit, show that the eccentricities agree, and compare a_1/a_2 with m_2/m_1. Explain how the trajectory plot and these measurements show that the center of mass is a shared focus rather than the center of either ellipse.
- Report the initial total momentum and include the center-of-mass deviation plot.

Part 3: Recovering the Fixed-Center Limit (35 points)

The fixed central mass in Assignment 6 was externally constrained to remain motionless. Here both bodies are allowed to move. Show that the interacting-body model approaches the fixed-center result when one body is much more massive than the other.

Use

$$m_{\text{heavy}} = 1 \qquad m_{\text{light}} = 10^{-6} \qquad d = 1$$

Complete **two separate simulations** with this mass ratio. The first simulation is an elliptical orbit and the second simulation is a circular orbit. Use the same preliminary positions in both simulations:

$$\mathbf{r}_{\text{heavy}}(0) = (0, 0) \qquad \mathbf{r}_{\text{light}}(0) = (d, 0)$$

Then call your center-of-mass-frame function before recording the initial states. The transformation is a change of reference frame, not a physical impulse applied to either body. It changes the individual velocities while leaving their relative velocity unchanged.

Simulation 1: Elliptical orbit

Initially leave the light body stationary and give the heavy body the velocity

$$\mathbf{v}_{\text{light}}(0) = (0, 0) \qquad \mathbf{v}_{\text{heavy}}(0) = (0, 0.8)$$

Use your function to transform these values to the center-of-mass frame. Report the preliminary and transformed velocities. Explain why the transformation gives the resulting speeds. The relative orbit should be bound and elliptical because its initial relative speed is below the circular speed. Its semimajor axis should satisfy

$$\frac{1}{a} = \frac{2}{d} - \frac{v_0^2}{G(m_{\text{heavy}} + m_{\text{light}})} \qquad \text{with} \qquad v_0 = 0.8$$

and its expected period is

$$T_{\text{ellipse}} = 2\pi\sqrt{\frac{a^3}{G(m_{\text{heavy}} + m_{\text{light}})}}$$

Simulate at least 10 expected periods.

Simulation 2: Circular orbit

Initially leave the heavy body stationary and give the light body the correct tangential linear speed:

$$\mathbf{v}_{\text{heavy}}(0) = (0, 0) \qquad \mathbf{v}_{\text{light}}(0) = (0, v_{\text{c}}) \qquad v_{\text{c}} = \sqrt{\frac{G(m_{\text{heavy}} + m_{\text{light}})}{d}}$$

Again use your function to transform the system to the center-of-mass frame before beginning the simulation. The expected period is

$$T_{\text{circle}} = \frac{2\pi d}{v_{\text{c}}}$$

Simulate at least 10 expected periods.

For both cases, compare the relative coordinate

$$\mathbf{r}_{\text{rel}} = \mathbf{r}_{\text{light}} - \mathbf{r}_{\text{heavy}}$$

with the corresponding fixed-center orbit from Assignment 6. Use $v_0 = 0.8$ for the elliptical comparison and the circular speed for the circular comparison. Do not compare the light body's position relative to the origin, since the origin is not the location of a fixed source in this model.

- Include trajectory plots showing the paths of both bodies for the elliptical and circular cases. Use a sufficiently tight view, or an inset, to make the heavy body's motion visible.
- Include a time-history plot of $K(t)$ for each body, $U(t)$ for each body, and $E(t)$ for each body as well as the whole system.
- Quantify the heavy body's maximum displacement from the center of mass in each case.
- Explain why both systems approach the fixed-center results.

Part 4: Three-Body Encounter and Ejection (45 points)

Use three equal masses and the following initial conditions:

$$G = 1, \qquad m_1 = m_2 = m_3 = 1,$$

$$\mathbf{r}_1(0) = (-0.5, 0), \qquad \mathbf{r}_2(0) = (0.5, 0), \qquad \mathbf{r}_3(0) = (0, 5),$$

$$\mathbf{v}_1(0) = \left(0, -\frac{1}{\sqrt{2}}\right), \qquad \mathbf{v}_2(0) = \left(0, \frac{1}{\sqrt{2}}\right), \qquad \mathbf{v}_3(0) = (0, -0.3).$$

Use your center-of-mass-frame function to adjust all positions and velocities before beginning the simulation. This system begins as a binary approached by a third body. Simulate long enough to observe the close interaction and the subsequent ejection.

Close encounters place strong demands on a fixed timestep. Repeat the simulation with smaller timesteps until the qualitative outcome is unchanged and the energy diagnostic is acceptable.

Required analysis

- Include a trajectory plot with all bodies identified consistently. Show a close view of the interaction and a wider view of the ejection if one set of axis limits cannot show both clearly.
- Include a time-history plot of $K(t)$ for each body, $U(t)$ for each body, and $E(t)$ for each body as well as the whole system.
- Include a plot of all three pairwise distances $r_{12}(t)$, $r_{13}(t)$, and $r_{23}(t)$.
- Report the minimum pairwise distance reached during the encounter and explain why it is relevant to the timestep choice.
- Identify the final bound pair. Support the identification with the pairwise distances and the pair energy
$$E_{ij} = \frac{1}{2}\mu_{ij}\|\mathbf{v}_i - \mathbf{v}_j\|^2 - \frac{Gm_im_j}{r_{ij}}, \qquad \mu_{ij} = \frac{m_im_j}{m_i + m_j}.$$
- Show that the remaining body is moving away from the center of mass of the final binary.
- Report the initial total momentum and include the center-of-mass deviation plot.

Extra Credit: A Small Mass at L_4 (20 points)

The triangular Lagrange point L_4 forms an equilateral triangle with two bodies in a circular orbit. Use
$$G = 1 \qquad m_1 = 1 \qquad m_2 = 10^{-3} \qquad m_3 = 10^{-6} \qquad d = 1$$
where d is the separation between the two larger bodies. Define
$$\mu = \frac{m_2}{m_1 + m_2}$$
and use the preliminary positions
$$\mathbf{r}_1(0) = (-\mu d, 0) \qquad \mathbf{r}_2(0) = ((1-\mu)d, 0) \qquad \mathbf{r}_3(0) = \left(\left(\frac{1}{2} - \mu\right)d, \frac{\sqrt{3}}{2}d\right).$$
The third body is therefore placed at L_4 relative to the two larger bodies. Use
$$\omega = \sqrt{\frac{G(m_1 + m_2 + m_3)}{d^3}}$$
and assign each body the linear velocity for counterclockwise rotation,
$$\mathbf{v}_i(0) = \omega\left(-y_i(0), x_i(0)\right).$$
For the stated positions, these preliminary velocities are
$$\mathbf{v}_1(0) = (0, -\omega\mu d) \qquad \mathbf{v}_2(0) = (0, \omega(1-\mu)d)$$
$$\mathbf{v}_3(0) = \left(-\frac{\sqrt{3}}{2}\omega d, \omega\left(\frac{1}{2} - \mu\right)d\right).$$
These velocities are tangent to the circular motion and have magnitudes $v_i = \omega r_i$. Call your center-of-mass-frame function after assigning all positions and velocities, then record the transformed initial states. Simulate at least 10 expected periods, where $T = 2\pi/\omega$.

- Include a trajectory plot showing all three bodies and the center of mass. Use a second tightly zoomed view if needed to show the motion of either larger body.
- Include a time-history plot of all three pairwise distances and determine whether the small body remains near L_4.
- Include time-history plots of $K_i(t)$, $U_i(t)$, and $E_i(t)$ for each body as well as $K(t)$, $U(t)$, and $E(t)$ for the whole system.
- Report the initial total momentum and include a center-of-mass deviation plot.
- Compare the measured rotation period with the expected value.

If you complete this extra credit, include a clearly labeled extra-credit subsection in your report. Describe this as a long-lived bounded numerical configuration rather than claiming that a finite simulation proves mathematical stability.

Short Report (20 points)

Prepare a concise report in LaTeX and submit the compiled PDF directly to Canvas. The report is an organized presentation of your work, not an additional lengthy assignment. Include your name, a descriptive title, all required figures, and concise captions or nearby explanatory text. Organize the discussion as a connected analysis rather than a numbered list of answers.
Address the following in a few sentences each:

- How the energy, momentum, period, and center-of-mass diagnostics validate the comparable-mass binary simulation.
- What the elliptical and circular large-mass-ratio cases reveal about transforming to the center-of-mass frame, why both approach the corresponding fixed-center results from Assignment 6, and why the interacting-body and fixed-center models remain physically different.
- How the three-body encounter redistributes energy, what evidence identifies the ejected body and final binary, and how timestep adequacy was checked.

Also include a brief note on testing or implementation choices that helped you verify your code and an **AI-use statement**. If you used AI, name the tool or tools, describe what they helped with, and explain how you checked or revised the resulting work. If you did not use AI, say so.

Submission

Submit all required Python source files and the compiled report PDF to Canvas. The figures belong in the report and do not need to be submitted separately.

Grading (150 points plus 20 points extra credit)

Part 1: Body class and N-body Velocity Verlet (15 points)

Category	What I look for	Pts
Body and history	Mass, state, histories, and state recording	5
Gravity	Correct pair distances and gravitational contributions	5
Verlet and diagnostics	Simultaneous updates and reusable system diagnostics	5
Total		15

Part 2: Comparable-mass binary (35 points)

Category	What I look for	Pts
Setup and CoM	Correct initial conditions and reusable CoM transformation	5
Trajectory	Equal-scale plot with initial positions and CoM	5
Energy	Individual and system $K(t)$, $U(t)$, and $E(t)$	5
Momentum and CoM	Initial momentum and tightly scaled CoM deviation	5
Period	Both measured periods and the expected value	5
Orbit measurements	Apsides, a_i, e_i, and mass-ratio test	5
Shared focus	Evidence that the CoM is the shared focus	5
Total		35

Part 3: Fixed-center limiting case (35 points)

Category	What I look for	Pts
Setups and CoM	Both setups, transformed values, and speed explanation	5
Periods	Expected periods and at least 10 periods per run	5
Trajectories	Both paths and heavy-body displacement	5
Fixed-center tests	Both relative-coordinate comparisons with Assignment 6	5
Energy	Individual and system energies for both runs	5
Momentum and CoM	Initial momentum and CoM deviation for both runs	5
Interpretation	Limiting behavior and physical distinction	5
Total		35

Part 4: Three-body encounter and ejection (45 points)

Category	What I look for	Pts
Setup and CoM	Correct initial conditions and CoM transformation	5
Timestep test	Smaller steps, acceptable energy, and no softening	5
Trajectories	Labeled close-encounter and ejection views	5
Energy	Individual and system $K(t)$, $U(t)$, and $E(t)$	5
Pair distances	All distances, minimum separation, and timestep link	5
Final binary	Identification supported by distances and pair energy	5
Ejected body	Motion away from the final binary's CoM	5
Momentum and CoM	Initial momentum and CoM deviation	5
Interpretation	Encounter outcome and energy redistribution	5
Total		45

Short report (20 points)

Category	What I look for	Pts
Organization	Title, connected structure, and readable presentation	5
Figures	All required figures with useful captions	5
Discussion	Accurate analysis supported by diagnostics	5
Verification and AI	Testing note and complete AI-use statement	5
Total		20

Extra credit: Small mass at L_4 (20 points)

Category	What I look for	Pts
Setup and CoM	Masses, L_4 state, linear velocities, and CoM frame	5
Orbit behavior	Trajectories, close view, and pair distances	5
Diagnostics	Individual and system energies, momentum, and CoM	5
Period and claims	Period comparison and appropriately limited claims	5
Total extra credit		20